Xinbo Xu (Dr)

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AREA OF INTERESTS

Indoor Built Environment, Thermal Comfort, Sleep Health, Personalized Model, Energy Consumption Simulation

CURRENT POSITION

- Research Fellow

Department of the Built Environment, National University of Singapore, Singapore 05/2025 to Present

EDUCATION

- Doctor of Engineering (Architecture, Building technology)

Department of Architecture, Shanghai Jiao Tong University, China 09/2019 to 03/2025

- Joint PhD student (Architecture, Building technology)

Center for the Built Environment, UC Berkeley, USA 01/2024 to 01/2025

- Bachelor of Engineering (Building environment and energy application engineering)

School of Energy Science and Engineering, Central South University, China 09/2015 to 06/2019

PUBLICATIONS

Note: An up-to-date publication list can be found on the Google Scholar profile (h index = 13).

[1] **X. Xu**, Z. Lian, Improve energy-efficient construction in China, *Science* 380(6648) (2023) 902.

[2] **X. Xu**, H. Zhang, S. Schiavon, Z. Lian, T. Parkinson, J.C. Lo, The effects of personal comfort systems on sleep: A systematic review, *Renewable and Sustainable Energy Reviews* 213 (2025) 115474.

[3] **X. Xu**, Z. Lian, Rethinking the calculation method of mean skin temperature in sleep research: Different formulas apply to different purposes, *Building and Environment* 251 (2024) 111231.

[4] **X. Xu**, Z. Lian, Which physiological measurements can characterize core and surface body temperature? A case study in stable thermal environment, *Building and Environment* 247 (2024) 111019.

[5] **X. Xu**, Z. Lian, Skin temperature for thermal sensation evaluation - is it valid everywhere?, *Building and Environment* 230 (2023) 110008.

[6] **X. Xu**, Z. Lian, A heat transfer model for sleep quality evaluation, *Building and Environment* 222 (2022) 109369.

[7] **X. Xu**, L. Lan, J. Shen, Y. Sun, Z. Lian, Five hypotheses concerned with bedroom environment and sleep quality: A questionnaire survey in Shanghai city, China, *Building and Environment* 205 (2021) 108252.

[8] **X. Xu**, H. Zhang, Z. Lian, H. Xu, Sex difference in body temperature and thermal perception during nighttime sleep: A time series analysis, *Energy and Buildings* (2025) 115995.

[9] **X. Xu**, Z. Lian, A sleep staging model based on core body temperature rhythm, *Energy and Buildings* 310 (2024) 114099.

[10] **X. Xu**, G. Wu, Z. Lian, H. Xu, Feasibility analysis of applying non-invasive core body temperature measurement in sleep research, *Energy and Buildings* 303 (2024) 113827.

[11] **X. Xu**, S. Li, Y. Yang, Z. Lian, Effects of thermal environment on body temperature rhythm and thermal sensation before and after getting into bed: A laboratory study in Shanghai, China, *Energy and Buildings* 301 (2023) 113748.

[12] **X. Xu**, J. Zhu, C. Chen, X. Zhang, Z. Lian, Z. Hou, Application potential of skin temperature for sleep-wake classification, *Energy and Buildings* (2022) 112137.

[13] **X. Xu**, W. Liu, Z. Lian, Dynamic indoor comfort temperature settings based on the variation in clothing insulation and its energy-saving potential for an air-conditioning system, *Energy and Buildings* 220 (2020) 110086.

[14] **X. Xu**, H. Zhang, G. Wu, Z. Lian, H. Xu, Sex differences in body temperature and thermal perception under stable and transient thermal environments: A comparative study, *Science of The Total Environment* 951 (2024) 175323.

[15] **X. Xu**, H. Du, Z. Lian, Discussion on regression analysis with small determination coefficient in human-environment researches, *Indoor Air* 32(10) (2022) e13117.

[16] **X. Xu**, Z. Lian, Objective sleep assessments for healthy people in environmental research: A literature review, *Indoor Air* 32(5) (2022) e13034.

[17] **X. Xu**, Z. Lian, J. Shen, T. Cao, J. Zhu, X. Lin, K. Qing, W. Zhang, T. Zhang, Experimental study on sleep quality affected by carbon dioxide concentration, *Indoor Air* 31(2) (2021) 440-453.

[18] **X. Xu**, L. Lan, Y. Sun, Z. Lian, The effect of noise exposure on sleep quality of urban residents: A comparative study in Shanghai, China, *Building Simulation* 16(4) (2023) 603-613.

[19] **X. Xu**, Y. Yang, T. Cao, T. Nie, Z. Lian, Four kinds of body temperatures and their relationships with thermal perception, *Journal of Thermal Biology* 114 (2023) 103600.

[20] **X. Xu**, Z. Lian, J. Shen, L. Lan, Y. Sun, Environmental factors affecting sleep quality in summer: a field study in Shanghai, China, *Journal of Thermal Biology* (2021) 102977.

[21] **X. Xu**, Z. Lian, Optimizing bedroom thermal environment: A review of human body temperature, sleeping thermal comfort and sleep quality, *Energy and Built Environment* 5(6) (2024) 829-839.

[22] **X. Xu**, S. Zhou, Z. Lian, Sleep staging based on skin temperature: A neural network prediction model, *Healthy Buildings 2023: Asia and Pacific Rim*, 2023.

ACADEMIC ACTIVITIES

Peer Reviewer of: *Sustainable Cities and Society*, *Building and Environment*, *Energy and Buildings*, *Journal of Thermal Biology*, *Science of the Total Environment*, *Indoor Environments*, *Indoor Air*, *Environmental Monitoring and Assessment*, *Indoor and Built Environment*, *Advances in Building Energy Research*, *AIMS Biophysics*, *Archives of Public Health*, *Eksploatacja i Niezawodnosc - Maintenance and Reliability*, and *5th International Conference on Building Energy and Environment*.

AWARDS

- National Scholarship for Doctoral Students, The Chinese Ministry of Education, China 12/2024
- National Scholarship for Doctoral Students, The Chinese Ministry of Education, China 12/2023
- National Scholarship for Doctoral Students, The Chinese Ministry of Education, China 12/2021
- Best Paper Award, 1st Chinese Conference on Thermal Comfort and Sleep Environment 07/2023
- Best Paper Award, 3rd Chinese Academic Forum on Thermal Comfort and Sleep Environment 07/2021

MAINLY PARTICIPATED PROJECTS

- Research and demonstration of key technologies for environmental monitoring, evaluation, and guarantee in healthy residential areas, No. 2022YFC3803201
  Funded by National Key R&D Program of China 10/2021 to 03/2025
- HEATS Project: Heat Exposure, AcTivity, and Sleep
  Funded by National Research Foundation of Singapore 01/2024 to 01/2025
- Effects of Ventilation in Sleeping Environments, ASHRAE project 1837-RP
  Funded by American Society of Heating, Refrigerating, and Air-Conditioning Engineers 09/2019 to 03/2021